



Affect, emotional intelligence and librarian-user interaction

Librarian-user
interaction

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Abstract

Purpose – The purpose of this paper is to explore practical ways in which librarians may better assist, understand and manage a library user's experience.

Design/methodology/approach – This paper is based on earlier work by Mills where 34 academics were interviewed on their information seeking behaviour. The concepts of affect and emotional intelligence have been introduced so information professionals can obtain a more clear understanding of the information environment.

Findings – In order to connect more closely with their user populations' information professionals could consider the following: embrace the key tenets of emotional intelligence as useful assistance strategies in user-librarian interaction; understand that personal interaction is important for many users; understand that such interaction can offer valuable insights into user understandings of the role of the library; understand that there is more to a library than resource access; understand that not all users share the same perceptions as librarians of the information values of such tools as catalogues and databases; appreciate that users see many roles for a library and these are individually constructed based upon past experience and current needs; extending the physical boundaries of the library into user communities is important for role development and accept that the key marketing strategy of commercial retailers to get customers to "buy" and return to buy, is relevant in environments such as libraries.

Originality/value – The paper builds upon research on the information seeking behaviour of academics and explores the idea that users select information sources for more than cognitive reasons, i.e. just to find out. The importance of the emotional aspect of user interaction with sources, including information professionals, in their search for information has been neglected. It is necessary to re-examine why and for what reasons users discriminate in their choice of information sources.

Keywords Emotional intelligence, Library users, Information profession, Libraries

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Introduction

Information professionals operate in a variety of information settings (public, private) and within these settings a myriad of information environments (academic library, school library, public library, organisation library). Perceptions exist, or images of these environments are held not only by those working in them but also by the users of them. It is a constant challenge for information professionals to create a responsive, relevant and resource rich environment. It is not enough for information agencies such as libraries to merely identify resources or to point the way to finding resources via catalogues, databases and the World Wide Web. Information service provision is constantly evolving and engaging in areas such as filtering and evaluating of sources, and in depth user education to promote an information literate community. With all of these developments there still remains a need to examine reasons for source choice, the information experience of users and to further evaluate the relationship between the information professional and the user. According to Dougherty (2001, p. 266) "a disconnect is developing between what libraries offer and what customers want". There is more to providing an information service than an ever-increasing range of resources. A closer investigation of the information environment of users and information professionals can provide some insights into what is needed. In



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discussing these issues this paper draws on research by Mills (2003a) into the information-seeking behaviour of university academics.

The information environment

The information environment of individuals and organisations plays a key role in influencing the types of interaction which may take place between information professional and user. Several factors influence this complex environment according to Renneker, Jacobson and Spink (2001), which are outlined below:

- perception of the environment;
- mental models of specific information sources;
- understanding of from where and from whom to seek information;
- the richness of the resources;
- affective responses to queries;
- experience with “information system feedback”;
- expectations of communication flow;
- previous experience with resources;
- previous experience with information providers.

The construction of the information environment for an information seeker is very individual and a perception of the “general” information environment is comprised of many of the above factors and their interaction. The environment and what creates it is so complex that it seems impossible to dissect and categorise the factors involved; all factors may be involved at a particular point in time and space and at other times only a few of the factors may be involved. Some factors such as users’ perception of the information environment seem to be embedded in the sub conscious and are hard if not impossible to easily understand (Mills 2003a).

Affective realm

A closer study of the factors outlined above reveals that many of them relate to what Julien (1999, p. 588) labels the affective realm, which can generally be said to describe the concepts of affect, preferences, evaluations, moods, and emotions where:

“Preferences” refer to subjective reactions, and are described on a scale of “unpleasant” to “pleasant”. “Evaluations” are straightforward reactions to other people, material objects, or experiences. “Moods” are less specific, and not directed towards specific external persons or objects. “Emotion” is a concept that includes the broad range of affects, and includes all the fine gradations of feeling that humans identify (Fiske and Taylor 1991 cited in Julien 1999, p. 588).

Most notable factors of interest in this paper are: perception of the environment, affective responses to queries, prior experience with resources and information providers and expectations of communication flow. The role of affect in influencing the direction information seeking will take and on information professional-user interaction has been noted by many in the professional literature (for example, Kuhlthau 1991; Mills 2003a; Mills 2003b; Taylor 1968; Julien 1999). Mills (2003a) in recent research on the information-seeking behaviour of university academics found, like Kuhlthau (1991), that the main forces driving searches for information lay in the affective realm.

One of the most important of these forces is prior experience, including the affective experience, which in information seeking plays an important role in influencing future action.

Prior experience and personal constructs

These “prior experiences” are influenced by what Kelly (1963) calls personal constructs to describe the ways in which humans make sense of their worlds. A user’s personal constructs about a source, use of technology or an institution give rise to motivating/demotivating factors in relation to the use of that source. The interaction of information seekers with sources of information influences how and whether a source will be used in the future. Interaction with institutions such as libraries and with information professionals also creates personal constructs, which influence future use.

Kelly’s personal constructs seem to be what Forgas (2001, p. 50) talks about as our memory representations about the world, or the many environments in which we exist, and the role of affect in organising and activating them and the “inseparable relation between emotion and other aspects of cognition”. In fact

Affective influences on memory have widespread consequences for the way people think and behave. This occurs because we can only make sense of complex events by calling on our memories and prior experiences to interpret them...Affect influences not only the content of cognition and behaviour (what we think and do), but also the process of cognition, that is, how individuals think. (Forgas 2001, p. 50, p. 54)

The role of affect in our day-to-day lives is pervasive; it plays an important part in influencing our routine and less routine day-to-day decision-making. It is no less important in our working lives and in the world of library and information management. Mills (2003a) suggests that the affective realm plays a pivotal role in motivating and/or demotivating information-seeking behaviour. It is both an influence on a user’s formation and changing of personal constructs (or memory representations) regarding libraries and librarians and of sources; it is a key influence on image formation whether that be a library, librarian or a source that is used.

Emotional intelligence

A related concept, which emphasises the importance of affect, is emotional intelligence or EI. EI includes:

the ability to monitor one’s own and others’ feelings and emotions, to discriminate among them and to use this information to guide one’s thinking and actions [and] the capacity to perceive emotions, assimilate emotion-related feelings, understand the information of these emotions, and manage them. (Sternberg 2001, p. 188)

Made popular by Daniel Goleman (1996) in his book *Emotional Intelligence: Why It Can Matter More Than IQ* “emotional intelligence has attracted the attention of the public because it suggests that emotions convey sensible meaning, which requires understanding’ (Mayer, Ciarrochi and Forgas 2001, p. xii). The concept has been argued and discussed at length by psychologists and it appears to have key applicability in the management area in debates over differences between management and leadership and whether the difference between a manager and a leader can be attributed to an understanding of emotional intelligence. The importance of this concept has been outlined by Mayer (2001, p. 24) who suggests that “if emotions convey information, then dismissing such information is done at the organization’s risk”.

According to Forgas (2001, pp. 62–63) emotional intelligence:

includes a degree of awareness of how affective states infuse our memories, thoughts, judgements, and interpersonal behaviours...Affect is thus likely to influence many relationship behaviours, group behaviours, organizational decisions, consumer preferences, and health-related behaviours, and EI necessarily involves knowing when and how these effects occur.

EI is also a creation of the affective realm but operates more on an interactive interpersonal level. In library and information management a greater understanding of the affective realm and concepts such as personal constructs, memory representations and emotional intelligence would seem to have relevance. A greater understanding of these concepts would especially appear to have application in the following areas:

- information professional and user interaction (see for example Eidson 2000);
- information professional to information professional interaction;
- interpersonal relations in the workplace.

The affective factors influencing the “information environment” outlined above, perception of the environment, affective responses to queries, prior experience with resources and information providers and expectations of communication flow, can also be thought of more generally as image constructions. Images or perceptions are created of environments and people and prior experiences and expectations of communication flow are image constructions placed on prior events. Images are personal or individual creations that are socially influenced by the environment around us (Mills 2003a).

Affect, image and the information professional

This affective response of individuals to the world around them, to sources and to image creation, is determined by the personal constructs of the individuals or to commonly held images, for example images held of information professionals. Affect influences choice of source, use of technology and creation of image. Image as indicated earlier is closely influenced by affect, which in turn influences choice of source and use of technology.

Over time an individual constructs a personal view, or image, of sources used to find information and of institutions such as libraries and librarians. The role of affect in this construction is also important. Nahl and Tenopir (1996) see that:

the affective domain pulls together into one powerful category the entire motivational and emotional involvement of searchers. Within information science, affect has been defined as “the continuous motivational energy provided by one’s intent, goal, purpose, use, as well as the emotional dynamic features that determine the quality of the search process, e.g., perseverance...paying attention to detail, frustration, hope, disappointment, excitement, disbelief, etc” (p. 277).

Affect influences the constructs of source and images. The meeting of the source, image and affect results in motivation or demotivation in information seeking and determines the strategies and shape information-seeking behaviour will take. Individuals have their own sets of personal constructs/images, many of which are in a constant state of change and reformulation as they are influenced by new experiences. But without motivation there can be no behaviour. The strength of motivation is shaped by the balance between motivators (such as a positive prior experience with an information source) and demotivators (such as a negative prior experience with an information source) that are attached to personal constructs (discussed more fully in Mills, 2003a; and Mills and Bannister, 2001). Motivators and

demotivators are vital ingredients in understanding the reasons for source selection towards the satisfaction of an information need and a successful approach to information-seeking behaviour.

Research findings

Research conducted by Mills (2003a) on the information-seeking behaviour of academics at a medium-sized university in Wagga Wagga, a rural town with a population of 60,000 people in New South Wales, Australia indicates that the affective realm is often at the root of choice of source and determines if the cognitive need will or will not be met. Information-seeking behaviour is very individual and has been personally constructed through a combination of motivating and demotivating influences. Academics in this research had their own unique approaches to their information-seeking behaviour fixed in their personal constructs. Each approach had been constructed and was in a continual state of reconstruction. The affective realm played a major role in creation of attitudes, feelings, beliefs, perceptions about technology, choice of source and image formation relating to source. For example one of the thirty-four participants interviewed in the research (participant 25) found some people in the library a "bit forbidding" so questions were not asked of them. The affective realm lies at the root of motivation to engage or not engage in information-seeking behaviour and is largely based in experience.

Experience and personal construct formation

Experience is embedded in personal construct formation and is the single most powerful determinant of the direction information seeking will take. Each academic has been exposed to a variety of experiences that make up individual information-seeking experiences and these influence the direction information seeking will take in the future. It is too simplistic to suggest that recent experience with technology is responsible for most of the changes to an academic's information-seeking behaviour over the past five years. An academic's subject discipline, for example, is another powerful predictor of what sources will be selected in information seeking. There are numerous studies documenting the different approaches of subject disciplines to information seeking and, within these disciplines, there are different approaches relating to specific subjects. There are other complicating factors such as the complexities of source selection in multidisciplinary studies, as Westbrook's (1999) work illustrates. The work environment of the academic interacts with the need to know, cognition, and motivation to initiate information-seeking behaviour.

Typically an academic in this research had pre-existing personal constructs relating to information seeking, built up over years of experience and changed by new external influences such as source use. Internal influences such as feelings, emotions, and attitudes towards these new experiences acted on existing personal constructs. The personal constructs formed, for example, about use of databases, then gave rise to motivators and demotivators. For example participant 22 found technology a problem. She was motivated to undertake database and other online searches with experts because as an independent searcher she thought she would invariably get stuck.

Images of library and librarian

Images of the library and librarian were created by many experiences and acted as motivators/demotivators influencing information-seeking behaviour. There was no one image identified of libraries or librarians. Images were often based on prior

experiences and perceptions of resources availability and accessibility, which were particularly found to be important in image construction and the resultant motivators/demotivators to using library services.

The vast majority of academics held positive images about the role of the librarian and library. There was no suggestion of the librarian as a “stern woman with a bun” a stereotype often portrayed in the literature. Importantly, participants identified a number of roles, summarised below. This is important because these roles encompass functions the librarian is generally thought to perform or which academics would like to see performed. They could be used as motivators relating to library use:

- librarian as keeper of the collection;
- librarian as creator of independent user;
- librarian as “knowledge navigator”;
- librarian as information technician;
- librarian as document deliverer;
- librarian as educator of academics;
- librarian as educator of students;
- librarian as interactive partner;
- librarian as information provider.

There was far from universal acceptance among participants about the degree of importance that they would ascribe to each role. There were also varying degrees of success attributed to the librarian’s ability to perform each role. Some roles attracted positive and negative feelings, as well as a diversity of views as to what should be the role as compared to what it was. It was a positive sign that participants identified key roles for librarians in this technological age. In terms of Mills’ (2003a) research there seems little reason to question Taylor’s (1968, p. 82) comments in his classic paper on information seeking. Taylor concluded that whether or not a librarian is asked for assistance is based upon factors such as “the inquirer’s image of the personnel, their effectiveness, and his previous experience with this or any other library or librarian”.

User’s images of self

At least one-third of academics interviewed indicated that they were concerned that they might appear to library staff as being unable to use, or having difficulty in using, the library. Some participants avoided library staff so as to not appear out-of-date or unable to navigate the OPAC or databases. This avoidance behaviour took many turns; for example browsing on the shelves was seen as non-threatening and potentially rewarding, and was engaged in preference to seeking advice from librarians about using the catalogues, especially if changes to the interface had occurred since the last time it was used. As Palmer (1996 cited in Spanner 2001, p. 359) noted “The potential for discovery was great enough to make browsing worthwhile. Browsing was essential to the gathering of information and the probing of new and peripheral areas”.

The negative image some participants held of librarians and/or libraries and the potential threat to their self-esteem when asking a question is another example of avoidance behaviour. Perceptions of being faced with information overload if library

staff were approached for assistance also led to browsing or often frustrating searches for information through the World Wide Web and databases. Whether or not library staff were approached for assistance also depended upon concepts of self efficacy held by respondents. In some cases, if self-esteem was not considered to be threatened by asking for help then it was requested, but protection of self-esteem was a powerful deterrent to information-seeking behaviour in the use of libraries. Research by Hull (2000) on experience of users in academic libraries in the UK corroborates these findings. Academics, while generally adept at finding what they want, nevertheless experience a range of problems when using libraries, the importance of which may well be underestimated by library staff. Problems for students in using libraries have been well documented by Mellon (1986), Onwuegbuzie and Jiao (1998) and Cleveland (2004). In relation to student use of academic libraries Smith (2004) says that "the predetermined attitudes brought to libraries by students may influence their response to library instruction. The ability to feel at ease in the library is determined by the library's environment, its staff and the individual student." Carlisle (2004), in discussing reference librarians, points out that they

...should be conscious of the fact that students come to the library with more than just a need for information. They also come with pre-conceived ideas about the library, including fears and apprehensions based on previous encounters or expectations. Those involved in academic reference services can better reach all students by providing reassurance that there is no need to feel embarrassed or ashamed about not understanding what is a complex and ever-changing academic library environment.

Like student users of academic libraries a typical academic library user probably does not exist; there are many types of users with varying degrees of information-seeking confidence. Supporting this finding Marie-Jose Griffiths (Hoffman 1999, p. 196) asks a question in a discussion on users and their use of information: "Given different information seeking patterns, can end users be spoken of as one category? How does one design information products that will be used by a wide range of people?" Stanley (2004, p. 16) refers to library users as either hunters or gatherers and moving between the two depending upon needs. Assumptions for service should not be made based upon a belief that there is a knowing, confident, experienced information seeker.

Changing roles-different images

Recent literature focuses upon the increasing need for librarians to play a role in interpreting, analysing, and selecting relevant information for their users. For example Griffiths (2000) suggests that "The creation of validated collections of digital materials and their relationship to validated non-digital materials will offer a significant value-add to the serious information seeker". Others, such as Tenopir (2001), refer to the increasing need to provide instruction in digital library and World Wide Web use to users. Although as Herring (2001, p. 217), suggests academics "do not consider the Web to be sufficient as a sole source to deliver the type or quantity of research information they need..." But if these tasks are to be carried out then there is a need for human interaction with users if the best use is to be made of the increasingly sophisticated and numerous means of access to information that are available. Many participants had feelings of inadequacy about searching and wanted assistance; for example participant 22 said: "I much prefer to do a search with an expert who I can call on when I get stuck as I invariably do...". Participant 26 felt changing technology was a problem for him in undertaking his information seeking:

that's also a bit of a hindrance...unless you are using information seeking very regularly, if you leave it for a few weeks you can be sure it's going to be different. You can never be confident in what you already know.

The information professions have a human services orientation and we know that people prefer to get their information from other people. But at the same time library services to users, especially to academics, are increasingly being seen in terms of remote access, often with potential for online user education rather than face to face instruction. A recent case for the continuation of physical library as place though has been presented by Choy (2005):

Physical libraries will continue to be important fixtures in the future campus for the simple reason that they are an important social and community space in addition to being depositories of books and other information objects. In a hybrid library where many media co-exist, space is essential to mediate between users, collection and services. Even when everything has gone completely electronic, we will still have to invent new spaces to cater to other aspects of learning. Learning is a social and multi-dimensional activity that relies on human interaction, multiple senses, chance encounters and serendipity for it to be effectively carried out. Libraries are special community spaces, with powerful symbolic and cultural connotations of learning, intellectual freedom and optimism. There is no more potent place to engage in study and learning.

American Library Association (ALA) *Guidelines for Behavioral Performance of Reference and information Service Providers* (2004), p. 1) acknowledge the shift away from "in-person encounters" with use of email and chat reference services but suggest that this move has not lessened the need for good communication skills:

In all forms of reference services, the success of the transaction is measured not only by the information conveyed, but also by the positive or negative impact of the patron/staff interaction. The positive or negative behavior of the reference staff member (as observed by the patron) becomes a significant factor in perceived success or failure.

The *Guidelines* (2004) identify three categories – general, in person and remote – that cover five main areas in interaction between reference librarians and users: approachability, interest, listening/inquiring, searching and follow up.

Human interaction with users need not always occur face to face. As the ALA *Guidelines* suggest the need for good communication skills remains whatever the type of reference encounter. The non-verbal and verbal cues evident in face to face interaction can be replaced by good communication skills in the virtual reference encounter.

Conclusion

The influence of affect in the success of librarian-user interaction is embedded in appropriate verbal and non-verbal communication skills.

While in the university environment, for example, academics still need to interact with each other to exchange ideas and to stimulate ideas, it is often asked if academics still need the assistance of librarians, still need to ask for assistance in satisfying their information needs. For the academics in the research reported in this paper it is suggested that some degree of interpersonal interaction between librarian and user is a given and is necessary. In the paper by Frank *et al.* an increase in this interpersonal interaction was proposed as a way to increase the meaning and relevance of the library. For example:

While traditional in-person visits to libraries continue and electronic information resources provide libraries and their respective clientele with innovative and effective access mechanisms, libraries cannot ignore the possibility, and in some cases reality, of becoming passive providers of information. The net result is that librarians need, and must embrace the opportunity, to engage scholars with innovative and effective services that stretch beyond traditional library boundaries. Essentially, the move to electronic services and information provision reduces the physical role of the central academic library and drives librarians to adapt library-bound services, transcend traditional boundaries, and move from reactive to proactive service models (Frank *et al.* 2001, p. 91).

In order to connect more closely with their user populations and avoid becoming “a shop with few customers”, information professionals could consider the following:

- accept that the concept of affect is influential in assisting successful librarian-user interaction;
- embrace the key tenets of emotional intelligence as useful assistance strategies in user-librarian interaction;
- understand that personal interaction is important for many users;
- understand that such interaction can offer valuable insights into user understandings of the role of the library;
- understand that there is more to a library than resource access;
- understand that not all users share the same perceptions as librarians of the information values of such tools as catalogues and databases;
- appreciate that users see many roles for a library and these are individually constructed based upon past experience and current needs;
- extending the physical boundaries of the library into user communities is important for role development;
- accept that the key marketing strategy of commercial retailers to get customers to “buy” and return to buy is relevant in environments such as libraries.

Librarians will need to find out more about what their users think of their roles and how they perform them. It is not enough to expect availability of resources, information filtering and user education to be the catalyst for library use, in person or remotely. It may be that too many assumptions have been made regarding the needs of users and the services users demand.

The potential power of the academic to obtain more information more rapidly, with little help from intermediaries, cannot be disputed. But academics need to be creative and a part of the development of this creativity is the need to be information seekers and to interact with those who provide information, including colleagues and librarians.

In the words of participant 21 a library needs to be seen as “a place whose value and meaning is defined by the users”.

References

- American Library Association (2004), *Guidelines for Behavioral Performance of Reference and Information Service Providers* (viewed 27 July 2006), available at: www.ala.org/ala/rusa/rusaprotools/referenceguide/guidelinesbehavioral.htm.

- Carlisle, H. (2004), "Literature review: the implications of 'library anxiety' for the provision of reference services in academic libraries", *INF501 Readings in Information Studies*, Wagga Wagga, Charles Sturt University.
- Choy, F.C. (2005), "Use of technology in Singapore libraries", *American Library Association Annual Conference*, Chicago, IL, March, American Library Association.
- Cleveland, A. (2004), "Library anxiety: a decade of empirical research", *Library Review*, Vol. 53 No. 3, pp. 177–85.
- Dougherty, R.M. (2001), "Being successful (nimble and agile) in the current turbulent environment", *The Journal of Academic Librarianship*, Vol. 27 No. 4, pp. 263–7.
- Eidson, M. (2000), "Using 'emotional intelligence' in the reference interview", *Colorado Libraries*, Vol. 26 No. 2, pp. 8–10.
- Fiske, S.T. and Taylor, S.E. (1991), *Social Cognition*, 2nd ed., New York, McGraw-Hill.
- Forgas, J.P. (2001), "Affective intelligence: the role of affect in social thinking and behaviour", in Ciarrochi, J., Forgas, J.P. and Mayer, J.D. (Eds), *Emotional Intelligence in Everyday Life: a Scientific Inquiry*, Philadelphia, PA, Psychology Press.
- Frank, D.G., Raschke, G.K., Wood, J. and Yang, J.Z. (2001), "Information consulting: the key to success in academic libraries", *The Journal of Academic Librarianship*, Vol. 27 No. 2, pp. 90–6.
- Goleman, D. (1996), *Emotional Intelligence: Why It Can Matter More Than IQ*, London, Bloomsbury.
- Griffiths, J.-M. (2000), "The roles of the information professions", available at: www.umich.edu/Bulletin/Feb-98/griffiths.html and www.asis.org/Bulletin/May-00/griffiths.html (viewed 27 July 2006).
- Herring, S.D. (2001), "Using the world wide web for research: are faculty satisfied?", *The Journal of Academic Librarianship*, Vol. 27 No. 3, pp. 213–9.
- Hoffman, D. (1999), "Users and their use of information", *Information Services Use*, Vol. 19, pp. 195–8.
- Hull, B. (2000), *Barriers Discouraging Access to Libraries as Agents of Lifelong Learning*, Research Report 31, Wetherby, West Yorkshire, Library and Information Commission.
- Julien, H. (1999), "Where to from here? results of an empirical study and user-centred implications for system design", in Wilson, T.D. and Allen, D.K. (Eds), *Proceedings of the Second International Conference on Research in Information Needs, Seeking and Use in Different Contexts, Sheffield, UK, 13–15 August, 1998*, London, Taylor Graham pp. 586–96.
- Kelly, G.A. (1963), *A Theory of Personality: the Psychology of Personal Constructs*, New York, Norton-Library.
- Kuhlthau, C.C. (1991), "Inside the search process: information seeking from the user's perspective", *Journal of the American Society for Information Science*, Vol. 42 No. 5, pp. 361–71.
- Mayer, J.D. (2001), "A field guide to emotional intelligence", in Ciarrochi, J., Forgas, J.P. and Mayer, J.D. (Eds), *Emotional Intelligence in Everyday Life: a Scientific Inquiry*, Philadelphia, PA, Psychology Press.
- Mayer, J.D., Ciarrochi, J. and Forgas, J.P. (Eds) (2001), "Emotional intelligence and everyday life: an introduction", in *Emotional Intelligence in Everyday Life: a Scientific Inquiry*, Philadelphia, PA, Psychology Press.
- Mellon, C.A. (1986), "Library anxiety: a grounded theory and its development", *College Research Libraries*, Vol. 47, pp. 160–5.
- Mills, J.J. (2003a), "Information-seeking behaviour of university academics", PhD thesis, Charles Sturt University.

- Mills, J.J. (2003b), "The affective dimension: a key motivator for implementing service quality", *Proceedings of 5th Northumbria International Conference on Performance measurement in libraries and information services*, Durham, UK, 28–31 July, the IFLA Section on Statistics.
- Mills, J.J. and Bannister, M. (2001), "Library and librarian image as motivators and demotivators influencing academic staff use of university libraries", *Performance measurement and Metrics*, Vol. 2 No. 3, pp. 159–71.
- Nahl, D. and Tenopir, C. (1996), "Affective and cognitive searching behavior of novice end-users of a full-text database", *Journal of the American Society for Information Science*, Vol. 47 No. 4, pp. 276–86.
- Onwuegbuzie, A.J. and Jiao, Q.G. (1998), "Understanding library-anxious graduate students", *Library Review*, Vol. 47 No. 4, pp. 217–24.
- Palmer, C.L. (1996), "Information work at the boundaries of science: linking library services to research practices", *Library Trends*, Vol. 45 No. 2, pp. 165–91.
- Renneker, M., Jacobson, A. and Spink, A. (2001), "Information seeking environment of a military university", *Preprint from Third International Conference on Information Seeking in Context, Goteborg, Sweden, 16–18 August, 2000*, pp. 1–15.
- Smith, L. (2004), Library Anxiety, implications for the Australian academic library: a literature review, *INF501 Readings in Information Studies*, Charles Sturt University, Wagga Wagga.
- Spanner, D. (2001), "Border crossings: understanding the cultural and informational dilemmas of interdisciplinary scholars", *The Journal of Academic Librarianship*, Vol. 27 No. 5, pp. 352–60.
- Stanley, J. (2004), "Hunters and gatherers: are you serving both their needs?" *inCite*, Vol. 25 No. 4, p. 16.
- Sternberg, R.J. (2001), "Measuring the intelligence of an idea: how intelligence is the idea of emotional intelligence?", in Ciarrochi, J., Forgas, J.P. and Mayer, J.D. (Eds), *Emotional intelligence in everyday life: a scientific inquiry*, Philadelphia, PA, Psychology Press.
- Taylor, R.S. (1968), "Question-negotiation and information seeking in libraries", *College Research Libraries*, Vol. 29, pp. 178–94.
- Tenopir, C. (2001), "Revelling in the changing world of reference librarians", In *Revelling in Reference 2001/RAISS-Reference and information Services Section Symposium Proceedings, 12–14 October, 2001, Melbourne*, Australian Library and information Association, Kingston, ACT, pp. 165–70.
- Westbrook, L. (1999), *Interdisciplinary Information Seeking in Women's Studies*, McFarland & Co., North Carolina, London.

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